AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_SINDEX **Submission:** model_2

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{6}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

An essentially complete solution following the markscheme approach. It sets up the encoding polynomial, gives the cyclotomic factorisations and their coprimality, correctly rules out index 6, establishes $N = 5$, and exhibits a valid multiset with maximum element 16, so $M = 16$. The setup omissions (why Q has no constant term, why the relevant LCM is the product of the cyclotomic factors) are standard enough to be trivial. The genuine defect is in the endgame: sharpening a bound of the form $r > 0$ to $r \geq 1$ requires the relevant coefficients to be integers, which is used but not justified (`ME2`), and the $M = 16$ and minimal-size arguments carry small gaps of the same kind. Complete with a minor error, so scores **6**.